

Final Project – Spring 2025

Project Summary Report:

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Fill the document as per the requirements of the project. You must include the following sections in the document:

1. High level description of the organization for which the network plan is being created.

This network design has been developed for **TechNet Solutions**, a mid-sized IT services company with operations spread across a central **Head Office (HO)** and three **branch offices**. The organization requires a reliable and secure communication infrastructure to support its internal operations across departments such as Administration, Sales, and IT. The primary goal of the network plan is to ensure seamless connectivity between all locations, centralized access to services like DHCP, DNS, HTTP, and Email, and simplified network management through proper segmentation and routing. The Head Office hosts all major network services, while each branch is equipped to access these resources over configured WAN links. The entire network design emphasizes stability, simplicity, and efficiency for day-to-day business needs.

2. Description of end devices.

The network consists of approximately **25 end devices**, distributed logically across all four locations. Each department in the Head Office is equipped with client **PCs** that are assigned IP addresses dynamically via a central **DHCP server**. Similar PCs are also present in each branch office. These client systems are used for accessing internal web resources, sending and receiving emails, and performing day-to-day tasks. Additionally, the network includes two **network printers**, one located in the Head Office and another in Branch 1, each configured with a static IP address and accessible within their local subnets.

Four centralized **servers** are hosted in the Head Office: a **DHCP server** that assigns IP addresses to end devices, a **DNS server** that resolves internal domain names (e.g., www.technetsolutions.com), an **Email server** for internal communication using configured user accounts, and a **Web server** that hosts the company's internal website. These servers are connected to the main floor switch in the HO and are assigned static IP addresses to ensure consistent service availability.

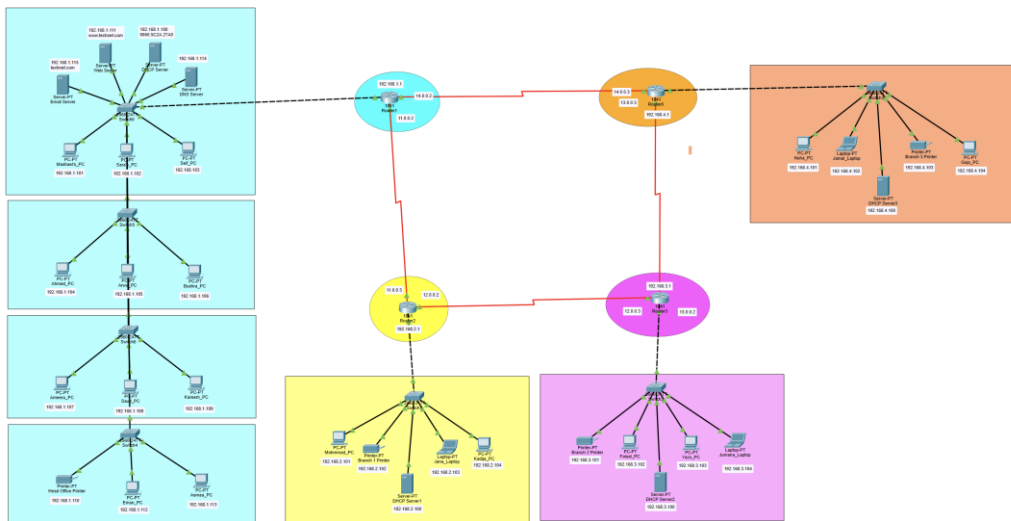
3. Description of intermediate devices.

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The network infrastructure uses several **intermediate devices** to ensure connectivity, segmentation, and communication between all offices. At the core of the network are four **routers**. The **R-HO** router in the Head Office acts as the central node and is connected to each of the three branch routers (**R-B1**, **R-B2**, and **R-B3**) via dedicated **serial WAN links**. These point-to-point links use private IP subnets and are configured with static routing to allow full network reachability between all office locations. This routing setup enables PCs and services in different subnets to communicate without the need for dynamic routing protocols.

The switching infrastructure includes **seven switches** in total. The Head Office is equipped with four switches—one for each floor—while each branch office has one switch to manage its local LAN. These **Layer 2 switches** connect end devices (PCs, laptops, servers, and printers) to their respective routers and manage local traffic within each office subnet. The Head Office switches also connect the core servers and allow inter-floor communication through trunk links where necessary.

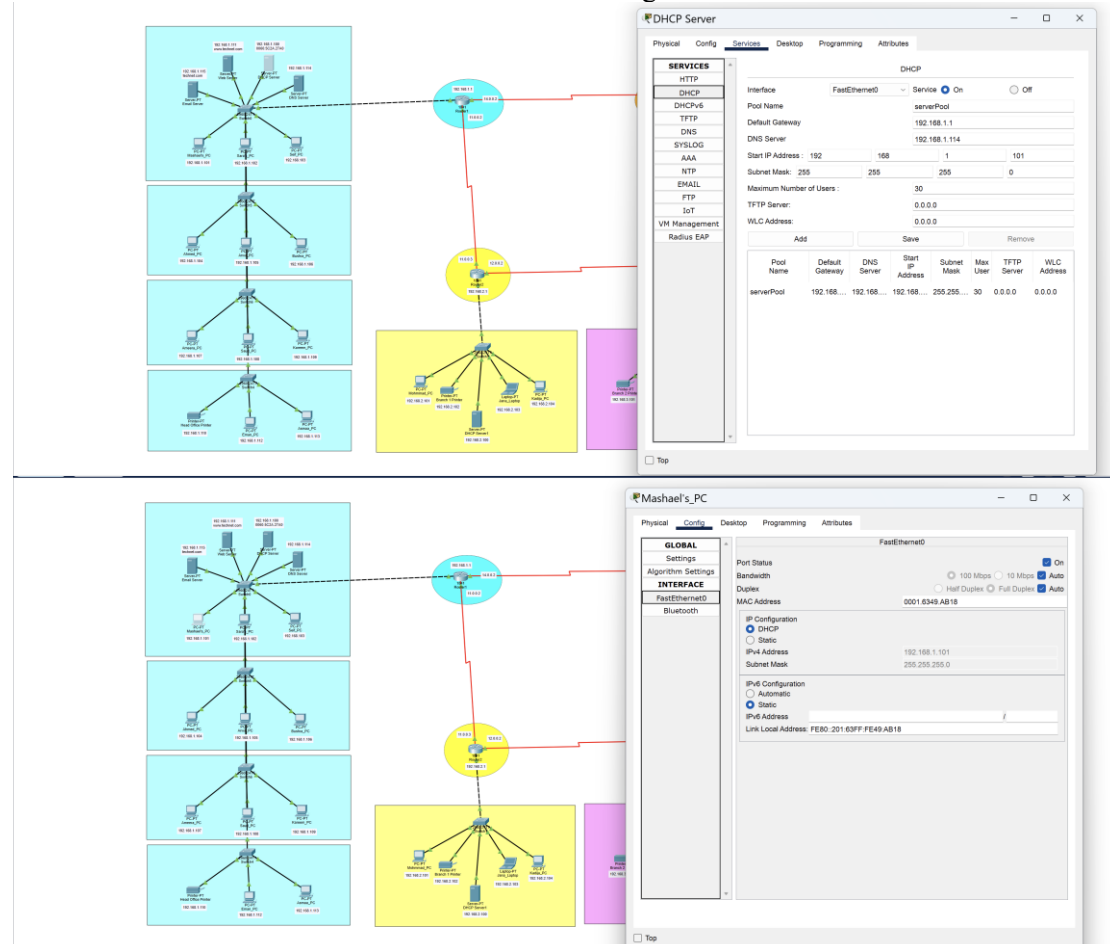
4. Screenshots of the completed network.



5. Screenshots demonstrating working of the network in packet tracer.

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This is the DHCP to confirm the IP addresses are assigned to the PC



The image displays a network diagram on the left, showing a central router (R1) connected to four switches (S1, S2, S3, S4). Each switch is connected to a group of PCs. The network is divided into two subnets: 192.168.1.0/24 and 192.168.2.0/24. On the right, two screenshots from the Packet Tracer interface are shown. The top screenshot is the 'DHCP Server' configuration window for the 'serverPool' interface. The bottom screenshot is the 'Mashael's PC' configuration window for the 'FastEthernet0' interface.

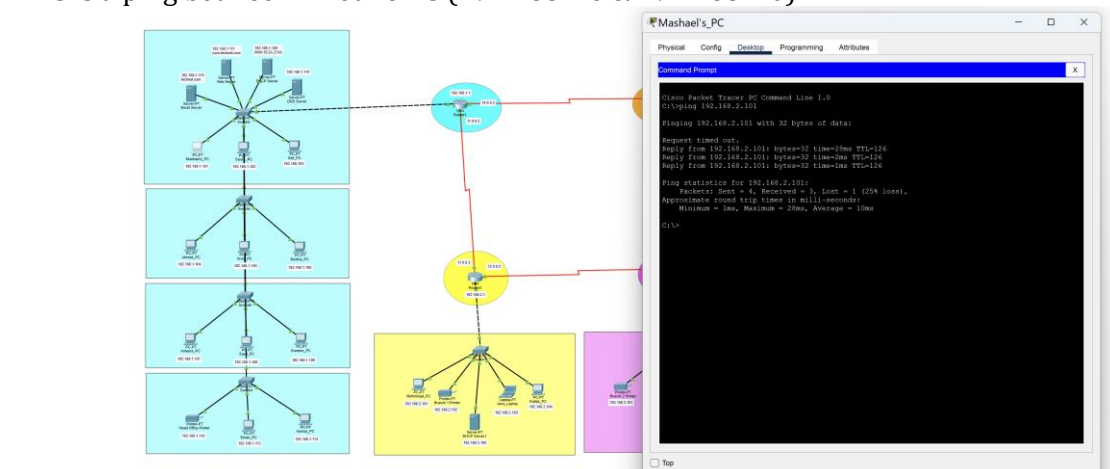
DHCP Server Configuration:

Interface	FastEthernet0	Service	On	Off
Pool Name	serverPool			
Default Gateway	192.168.1.1			
DNS Server	192.168.1.114			
Start IP Address	192.168.1.101			
Subnet Mask	255.255.255.0			
Maximum Number of Users	30			
TFTP Server	0.0.0.0			
WLC Address	0.0.0.0			

Mashael's PC Configuration:

Port Status	100 Mbps	10 Mbps	Auto
Bandwidth			
Duplex			
MAC Address	0001.6348.AB18		
IP Configuration			
Static			
IPv4 Address	192.168.1.101		
Subnet Mask	255.255.255.0		
IPv6 Configuration			
Automatic			
IPv6 Address			
Link Local Address	FE80::201:63FF:FE48:AB18		

This is a ping between 2 networks (192.168.1.0 & 192.168.2.0)



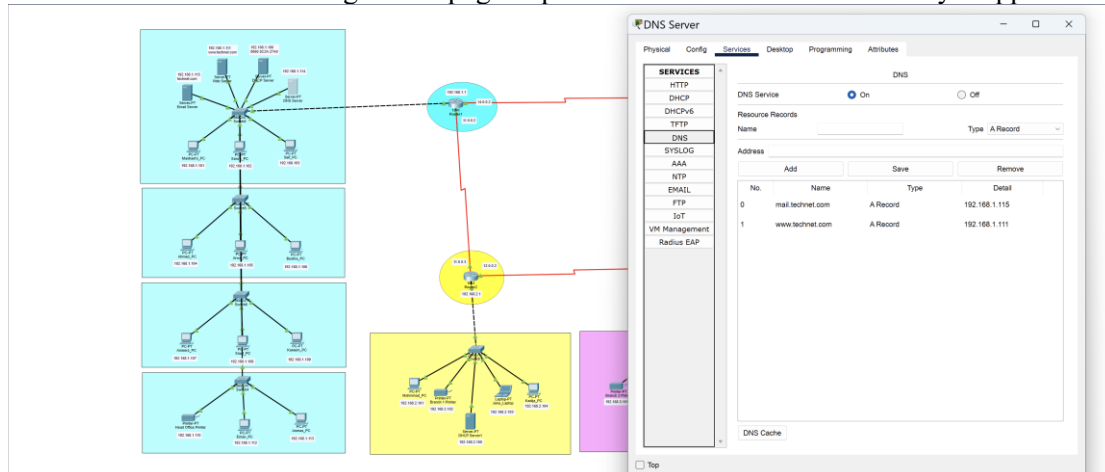
The image displays the same network diagram as above. On the right, a screenshot of the 'Mashael's PC' Command Prompt window shows the results of a ping command from 192.168.1.101 to 192.168.2.101.

Command Prompt Output:

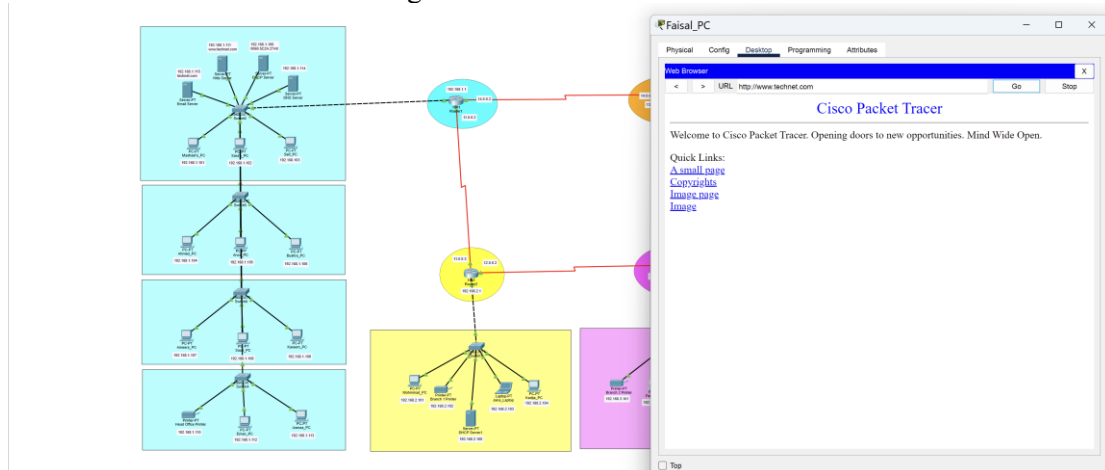
```
C:\Users\Mashael>ping 192.168.2.101
Pinging 192.168.2.101 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.101: bytes=32 time=124ms TTL=124
Reply from 192.168.2.101: bytes=32 time=124ms TTL=124
Reply from 192.168.2.101: bytes=32 time=124ms TTL=124
Ping statistics for 192.168.2.101:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milliseconds:
        Minimum = 124ms, Maximum = 124ms, Average = 124ms
C:\Users\Mashael>
```

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This is the DNS server configuration page to prove domain names are correctly mapped

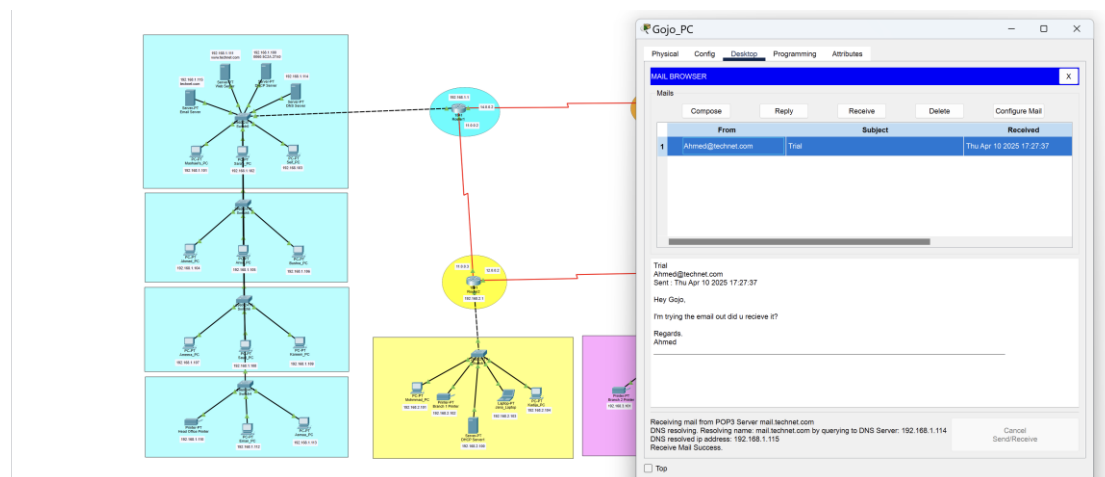


This is the website accessed through the DNS and web server

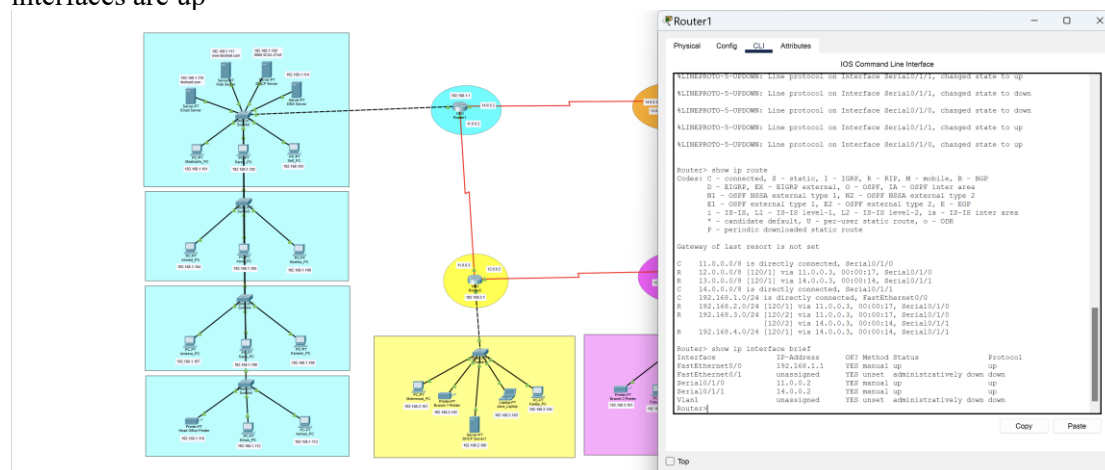


This is the email that has been accessed through the DNS and email server

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This is the routing table for router 1 that proves static routes are configured and serial interfaces are up



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This document and simulation file fulfill the requirements outlined in the grading rubric by providing a detailed description of the organization, structured explanation of end and intermediate devices, and functional demonstration of network performance. All design decisions are aligned with best practices and ensure the network is secure, maintainable, and scalable. No additional features were added beyond scope, keeping the setup practical and appropriate for the course objectives.